

Archaeologist

Kaplan idea → robot IL env/object laws · not an OXE replacement

Genealogy one-liner

Kaplan/Henighan LLM-CV scaling laws → (method analogy) → this paper's data-axis laws for single-task IL → efficient UMI collection + Diffusion Policy + ACT temporal ensemble → contrast OXE (multi-emb) / DROID (scene co-train) → informs Octo/OpenVLA/ π_0 mix thinking and BEHAVIOR per-skill design.

Lineage chain

CLEAR citations in paper + prior packs

Kaplan / Henighan 2020

Power-law method
on scale axis

UMI Chi'24

Handheld collect
→ robot deploy

DP Chi'23 + ACT TE Zhao'23

Policy + chunk
stitch

This paper Lin'25 Oral

Env/object laws
32×50 recipe

OXE / DROID contrast

Different goals
foundation / scenes

Octo / OpenVLA / π_0 / BEHAVIOR

Mix thinking
per-skill floor

Intellectual move

What changed when scaling laws entered robot IL

From tokens/params

To #envs / #objects / #pairs as the primary $X - Y = \text{optimality gap } 1 - S$, not CE.

From “more demos”

To diversity-first collection with an explicit $K \approx 50$ plateau — opposite folklore.

From foundation mix

To single-task zero-shot within category — complementary to OXE, not competing.

From vibes diversity

To fitted power laws + actionable 32×50 recipe verified on held-out tasks.

CLEAR vs TENTATIVE ancestry

Do not launder analogies into measured equalities

Kaplan et al. 2020	Method analogy (power law on scale)	CLEAR cite	Exponents NOT transferable
Hoffmann/Chinchilla	Compute-data optimality	Not fitted here	TENTATIVE analogy only
UMI 2402.10329	Collection + deploy interface	CLEAR	Inherited stack
Diffusion Policy	Policy family	CLEAR	DP-only limitation
ACT 2304.13705	Temporal ensemble only	CLEAR	Not ACT-as-primary
OXE 2310.08864	Related data scaling; X-emb FT goal	CLEAR contrast	Not replacement
DROID 2403.12945	Scene diversity prior	CLEAR cite	Complementary
Octo / OpenVLA / π_0	Generalist scale	Cited related	Stops before task axis

Stack inheritance — UMI + DP + ACT TE

This paper's systems spine is borrowed, then measured

UMI

Hand-held grippers
in-wild envs/objects
~90% SLAM-valid

Collection mechanism

Diffusion Policy

CNN 1D U-Net
DDIM-16 · hor 16
DINOv2 ViT-L FT

Policy family measured

ACT TE

Exp-weighted
8 steps · -0.01
stitch chunks @5 Hz

Smoothing only — CLEAR

Contrast — OXE / DROID / this paper

Same “data scaling” phrase · different X and goal

Pole	Scales	Primary claim	Relation
This paper	Envs × objs × demos	Power-law OOD; 32×50	Measured here
OXE	Robots × labs × skills	Positive X-emb transfer	CLEAR contrast
DROID	Scenes (one Franka)	Scene diversity co-train	CLEAR complement
UMI	Interface + locations	Wild without robot fleet	Mechanism used
Octo/OpenVLA/ π_0	Multi-task + compute	Generalist priors	Stops before task axis

Not an OXE replacement

Say this out loud in discussion

Discipline

Lin'25 answers “how many scenes/objects per skill for ~90% ZS?” — it does NOT answer “how to mix embodiments / labs / languages for a foundation policy.”

OXE remains the multi-embodiment aggregation story. DROID remains the same-robot scene-churn co-train story. This paper is the per-skill diversity meter.

Together

Use OXE for width · DROID for scene protocol · Lin'25 for per-primitive floor (32×50) · UMI for collection bandwidth.

What stuck — numbers that entered the syllabus

The Archaeologist's sticky-note set

diversity $\gg$ depth

moral of Figs 2-4,7

32 $\times$ 50

efficient frontier

$\alpha \approx -0.5 \dots -0.8$

Fig.5 exponents

K $\approx$ 50 plateau

demo saturation

ViT $\succ$ U-Net

capacity asymmetry

frozen / LfS fail

FT indispensable

$>40k$ / $>15k$

study scale

1191 pairs

0.99 prediction

Fig. 5 — the Kaplan-method hinge

Same log-log ritual · different X (envs/objects/pairs) · different Y (1-S)

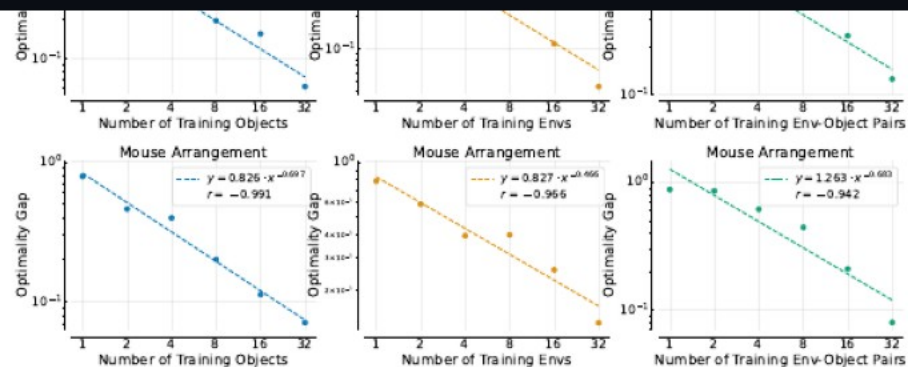


Figure 5: **Power-law relationship.** Dashed lines represent power-law fits, with the equations provided in the legend. All axes are shown on a logarithmic scale. The correlation coefficient (Pearson's r) indicates a power-law relationship between the generalization ability and the number of objects, environments, and environment-object pairs. See Appendix G.1 for data scaling laws on MSE.

ing further emphasizes that expanding the diversity of environments and objects is more effective than merely increasing the number of demonstrations for each individual environment or object.

4.2 POWER-LAW FITTING AND QUANTITATIVE ANALYSIS

We next explore whether our experimental results follow power-law scaling laws, as seen in other domains. Specifically, if two variables Y and X satisfy the relation $Y = \beta \cdot X^{\alpha}$, they exhibit a power-law relationship. Applying a logarithmic transformation to both Y and X reveals a linear relationship: $\log(Y) = \alpha \log(X) + \log(\beta)$. In our context, Y represents the optimality gap, defined as the deviation from the maximum score (i.e., $1 - \text{Normalized Score}$), while X can denote the number of environments, objects, or demonstrations. Using data from our previous experiments with a 100% fraction of demonstrations, we fit a linear model to the log-transformed data, as shown in Fig. 5. Based on all the results, we summarize the following data scaling laws:

¹Although we recognize the irreducible errors Y_{∞} associated with scaling the data alone, fitting a three-parameter model $Y = \beta X^{\alpha} + Y_{\infty}$ does not seem statistically justified given that we have only 6 data points.

+ / – learned since publication

What later work should keep / discard

+ Keep

- Diversity-pair budget as unit
- $K \approx 50$ plateau discipline
- Blind staged scoring
- Nested subset design
- Encoder FT + size over U-Net size

– Discard / nuance

- Demo-count as primary X
- MSE as ship metric
- Mean SR without env strat
- Silent DP-only transfer
- Treating 1191 as measured

Open question for successors

Do ACT / π_0 -FM / VLA heads move the exponents? Does M_{robot} enter the same power family as M_{env} ?
Task-count laws still missing.

Numbers tattoo — pin these

Every deck carries the same spine numbers

diversity $\gg$ depth

envs/objs/pairs $\gg$ K

32 $\times$ 50

pairs $\times$ demos $\rightarrow$ $\sim$ 90%

85-92.5%

Tab.1 SR $\cdot$ 4 tasks

$\alpha \approx -0.47...-0.84$

Fig.5 exponents

0.66 / 0.81 / 0.90

ViT-S / B / L

0.88 / 0.90 / 0.83

U-Net S / base / L

0.00 / 0.03

frozen / LfS fail

K $\approx$ 50

plateau / pair

>40k / >15k

demos / rollouts

1191 pairs

MA score 0.99 extrap.

DP + ACT TE

@5 Hz $\cdot$ horizon 16

FT 0.90 / LoRA 0.72

full FT required

Exam bite — Archaeologist

EXAM BITE

LLM scaling idea → robot IL env/object laws (Lin'25); collection via UMI; DP+ACT TE stack;
CLEAR contrast to OXE/DROID — not an OXE replacement. Kaplan = method analogy; exponents
TENTATIVE if claimed equal.

Archaeologist — close

Takeaways + exam bite

1. Kaplan → method analogy → Lin'25 env/object power laws ($Y=1-S$).
2. Stack: UMI collect + DP policy + ACT TE — CLEAR inheritance.
3. Contrast OXE (robots) / DROID (scenes) — complementary, not replacement.
4. Tattoo: diversity $\gg$ depth · 32×50 · Fig.5 α · CLEAR vs TENTATIVE exponents.

EXAM BITE

LLM scaling idea → robot IL env/object laws (Lin'25); UMI collect; not an OXE replacement.